

Week 6

Linear Systems of Equations and Phase Portraits

Tips and Extra Bits and Bobs

Introduction

Hey all here's just a few tips/helpful diagrams for this week's content.

Content which exists mostly just as shortcuts/verification in working out is listed as “extra” in the slide.

Extra: Nothing here is required knowledge!

A quick way to compute eigenvalues (2 by 2):

Hi all, if you wanted a quick way to compute eigenvalues for 2 by 2 matrices I would like to recommend this video here:

https://youtu.be/e50Bj7jn9lQ?si=SOVmiD_73GN1eB9h

In short, the gist of the algorithm is:

(1) Compute Trace A (Sum of diagonal of your matrix) and Det A

(2) Let $m = \text{Tr } A/2$, $p = \det A$

(3) $\lambda_{1,2} = m \pm \sqrt{m^2 - p} \rightarrow \text{Your eigenvalues!}$

Solving systems of linear equations

(1) Convert Differential Equations into matrix form

(2) Evaluate Eigenvalues and Eigenvectors of Differential Equation Matrix

(3) Evaluate solution based on Eigenvalue type:

Real, Distinct
Eigenvalues

$$y(t) = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} e^{\lambda_1 t} + \begin{bmatrix} v_3 \\ v_4 \end{bmatrix} e^{\lambda_2 t}$$

Repeated Real

$$y(t) = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} e^{\lambda t} + \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} e^{\lambda t} + t \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} e^{\lambda t}$$

Where u_1, u_2 is the solution of:

$$y(t) = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \operatorname{Re}(e^{\lambda_1 t}) + \begin{bmatrix} v_3 \\ v_4 \end{bmatrix} \operatorname{Im}(e^{\lambda_2 t})$$

Note: You determine the complex form by putting your complex eigenvalues in the form equivalent to Real, Distinct Eigenvalues.

You can then use exponent laws and Euler's Formula to expand out and separate your equation into real and imaginary components. (See this week's tutorial solutions)

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

(Where a, b, c, d are the coefficients of your matrix form)

Tips for drawing phase portraits

(0) Finding Critical Points (i.e. where: $(\dot{x}, \dot{y}) = (0,0)$ /where there is no change in the phase portrait) (For Non-Linear Phase Portraits)

(1) Sketching Eigenvectors – Eigenvectors corresponding to straightline paths in your phase portrait.

(2) Evaluating Slope on axis' – once you've managed to find your slope equations, you can evaluate $x = 0$ or $y = 0$, to find the slope on x/y-axis.

(3) Sketching Nullclines, for your phase portrait you can find where the velocity vectors are entirely horizontal/vertical by finding evaluating where $x = 0$ or $y = 0$ for your original differential equations.

(4) Evaluating Velocity Field Points – If you're unsure about the direction of a certain part of a plot or want some better resolution connecting two parts of your plot, you can always find an individual point in the velocity field by evaluating $(\dot{x}, \dot{y})$ at some coordinate (x, y) .

Extra!

Secret extra phase portrait drawing knowledge

(Image Source: Whitman College M244)

Poincare - diagram: Classification of phase portraits in $(\det A, \text{Tr } A)$ -plane

If I'm going to be honest, whenever someone says to me "draw a phase portrait" the following picture comes up:

This picture shows you all the different phase portraits you can come across when studying a system of linear differential equations.

Effectively you can determine what your phase portrait is going to look like based off of the determinant and trace of the matrix you are analysing.

For example, if you had the following matrix (from your system of equations):

$$A = \begin{bmatrix} 2 & 1 \\ -4 & 8 \end{bmatrix}$$

As:

$$\text{Tr } A = 10$$

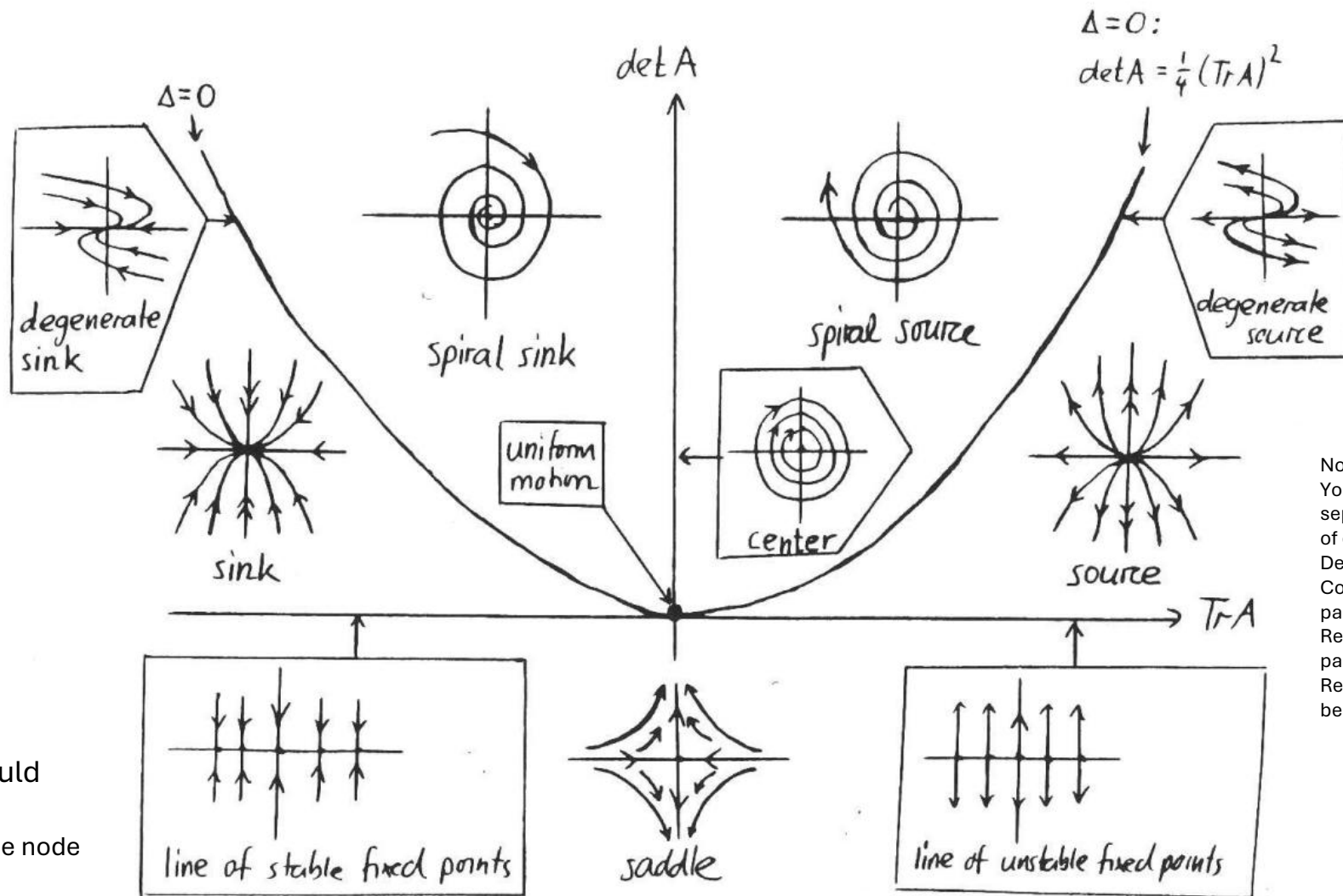
$$\text{Det } A = 20$$

And:

$$\lambda_{1,2} = 5 \pm \sqrt{5} \text{ (See note)}$$

You would have:

A Source node



Note:
You can also separate by type of eigenvalue for $\text{Det } A > 0$.
Complex, above parabola.
Repeated, on parabola.
Real, Distinct – below parabola

Alternatively, you might prefer to know the following heuristic:

If $\det A > 0$:

If you have two distinct real eigenvalues (source/sink node)

If you have one repeated eigenvalue (degenerate source/sink node)

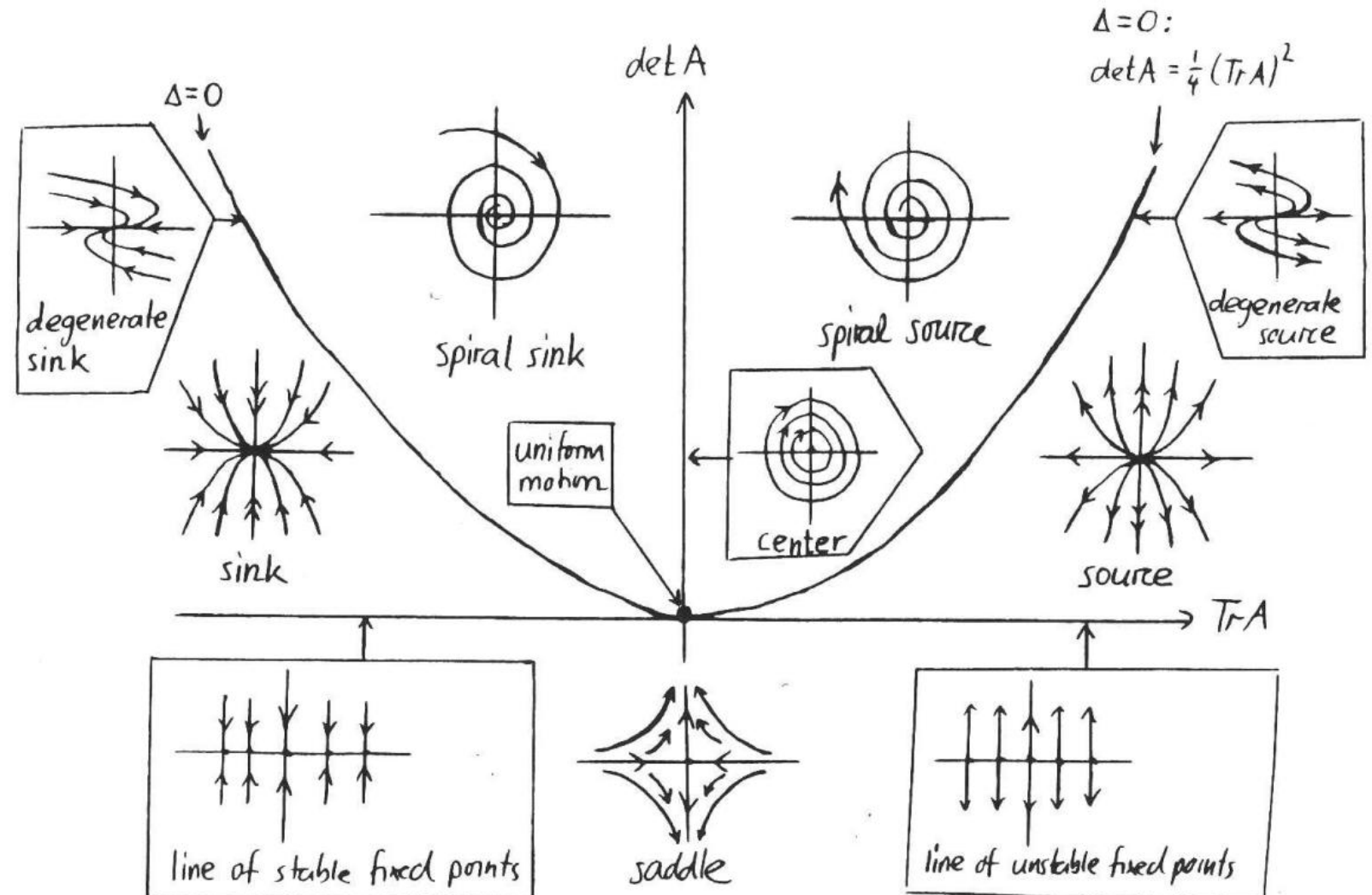
If you have complex eigenvalues (something with rotation)

Otherwise for $\det A < 0$:

Saddle point (one stable direction, one unstable direction)

You can then use $\text{Tr } A$, to check for stability/precise portrait type.

Poincaré-diagram: Classification of phase portraits in $(\det A, \text{Tr } A)$ -plane



Very unnecessary information at this point – but in case you would like some information on application.

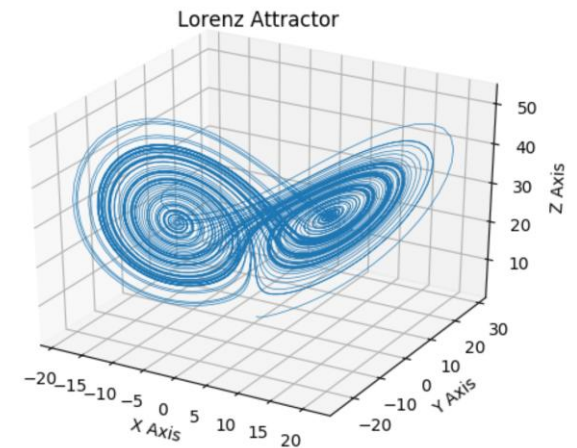
Why do we care?

I can say this in two senses.

- (1) The literal: For many of you, you will have a class in control theory/systems modelling sometime in the future. For these classes being able to “control”/manage stability of a dynamical system is an aspect of what you’ll likely be working on here.
- (2) Numerical issues, when working with differential equations most packages are good at numerically integrating solutions, however, quite a few don’t have the functionality to tell you about stability. With these tools from this week (and the next) you should hopefully be able to have some of the tools to analyse stability.

Finally, why do we look at only 2D here?

The honest answer is in 2D where we can pretty much describe any (even non-linear) system near critical points as a combination of the phase portraits on the previous slide. Notably, however, things get a lot more complicated in 3D, to an extent that you can end up with chaotic behaviour occurring (see Lorenz attractor) in your dynamical system.



(Image Source: Matplotlib, Lorenz Attractor)